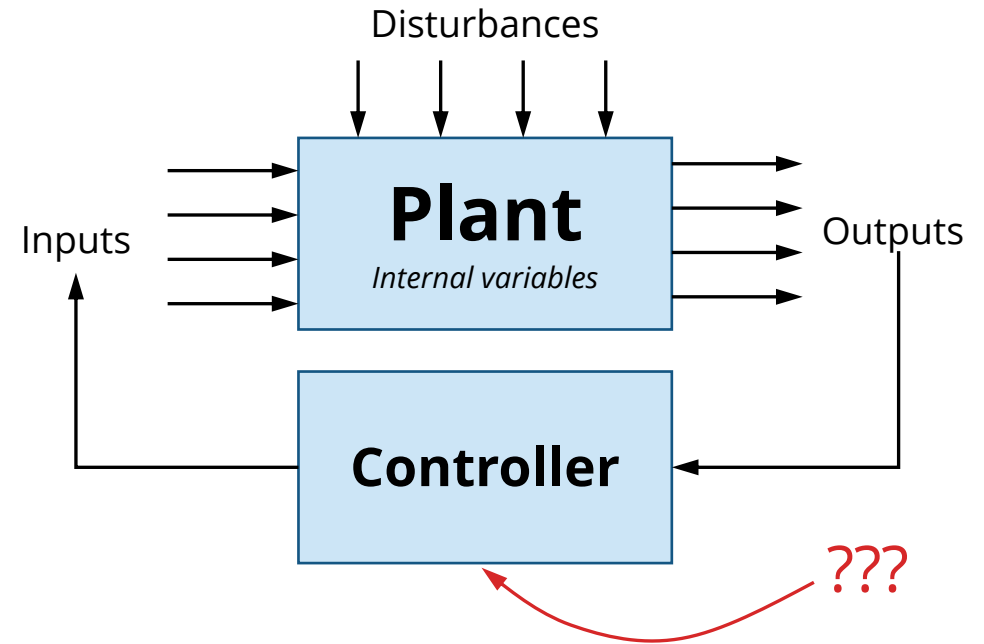




# AA 513 Multivariable Control

Lecture 05 // Week 5

Instructor: Prof. Karen Leung

# Announcements

- Exercises 4, 5, 6, 7, 8 due
- Next week, have (optional but strongly encouraged) check-in sessions by appointment to discuss research papers and tutorial topics.
  - Tentatively Wednesday evening
  - More details to follow.
- Research paper presentation
  - Follow PACES framework
- Tutorial
  - Go through a worked example or derivation. Support with code/plots
  - Apply material from course but on your own system
  - Tweak parts of an exercise and talk about it
- Guest lecture today! Dr Liam Kruse

# Last time

- Wrap-up Control Barrier Functions (CBFs)
- Principle of optimality

*When following an optimal solution, the remainder of the solution must also be optimal from that state onwards.*
- Value function  $V_{\pi}(x, t), u = \pi(x, t)$ 
  - Represents the *cost-to-go* from current state and time until end of horizon while following a **policy**
  - Compute the value function using **dynamic programming**
- Optimal value function  $V^*(x, t), u^* = \pi^*(x, t)$ 
  - Apply dynamic programming, but choose the *best* control input (solve an optimization problem at each step)

# Last time (discrete time)

- Consider cost  $g_N(\mathbf{x}_N) + \sum_{t=0}^{N-1} g(\mathbf{x}_t, \mathbf{u}_t, t)$  and dynamics  $f(\mathbf{x}_t, \mathbf{u}_t, t)$

- **Bellman equation** (deterministic dynamics, finite horizon)

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}_t \in U(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t + 1)$$

- **Stochastic dynamics**  $\mathbf{x}_{t+1} \sim p(\cdot | \mathbf{x}_t, \mathbf{u}_t, t)$

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}_t \in U(\mathbf{x}_t)} \mathbb{E}_{\mathbf{x}_{t+1} \sim p(\cdot | \mathbf{x}_t, \mathbf{u}_t, t)} [g(\mathbf{x}_t, \mathbf{u}_t, \mathbf{x}_{t+1}, t) + V^*(\mathbf{x}_{t+1}, t + 1)]$$

$$= \min_{\mathbf{u}_t \in U(\mathbf{x}_t)} \sum_{\mathbf{x}_{t+1} \in X} (g(\mathbf{x}_t, \mathbf{u}_t, \mathbf{x}_{t+1}, t) + V^*(\mathbf{x}_{t+1}, t + 1)) p(\mathbf{x}_{t+1} | \mathbf{x}_t, \mathbf{u}_t, t)$$

(Definition of expected value)

- **Infinite horizon** (Need to ensure the value is finite despite an infinite sum)

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}_t \in U(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + \gamma V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t + 1)$$

(Add a discount factor  $0 < \gamma < 1$  that discounts future costs)

# Practical considerations

- $V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}_t \in U(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t + 1)$ 
  - An optimization problem within the Bellman equation (+variants)
- For (small) discrete state and control spaces, and discrete time, we can enumerate over all possible controls to solve optimization problem
  - Store solution as a big table, indexing in state and time to obtain value and optimal control (exercise 9)
- For continuous state and control, we cannot enumerate
  - Require a more sophisticated optimization problem
  - The functional form of  $g(\mathbf{x}_t, \mathbf{u}_t, t) + V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t + 1)$  may not be preserved, i.e.,  $V^*(\mathbf{x}_t, t)$  has different functional form.
- Perform “**approximate** dynamic programming” and **function approximators** for value function

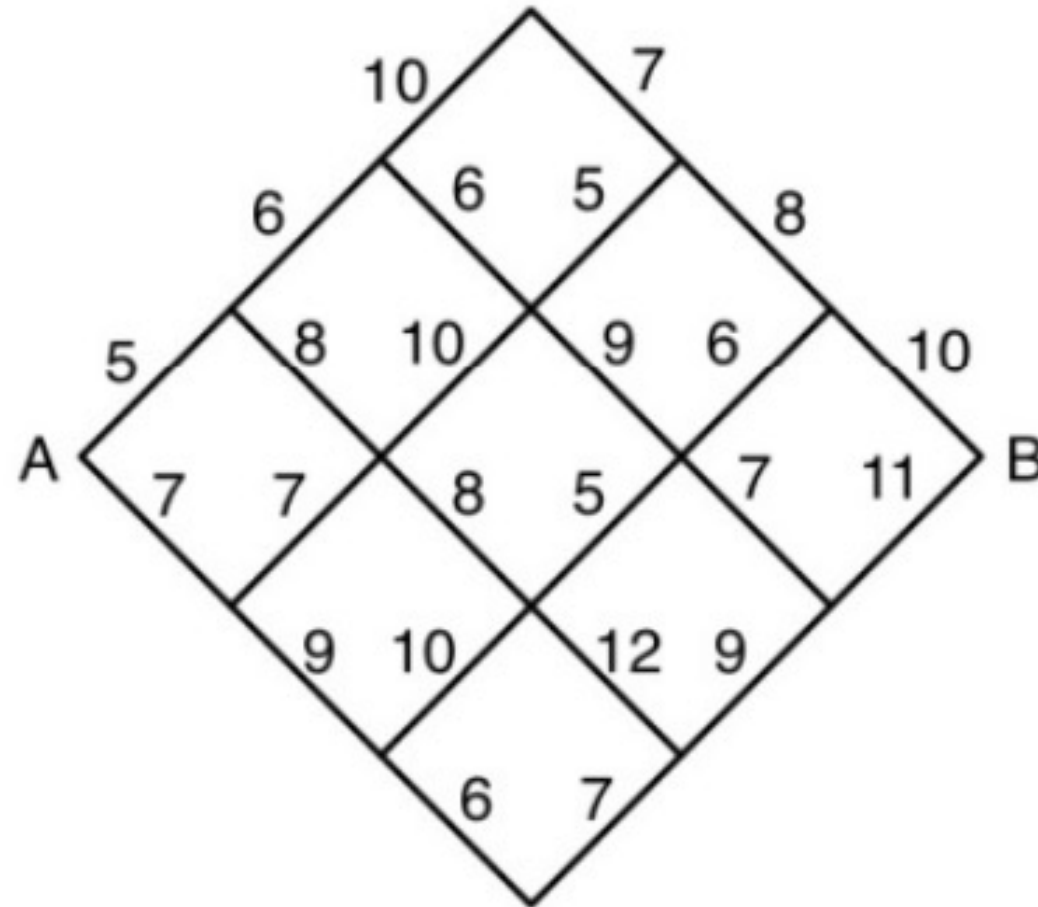
# This week

- Guest lecture: Dr Liam Kruse
- Continuous time: Hamilton-Jacobi-Bellman equation
- HJ Reachability
- Start of Linear quadratic regulator

# Exercises

- Ex9 Stochastic Bellman equation

# One more DP example...





# Continuous-time, continuous state & control

- Consider cost  $J_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$
- Cast as discrete time problem with timestep  $\epsilon$ 
  - $J_{discrete}(\mathbf{x}_t, \mathbf{u}_t, t) = \int_t^{t+\epsilon} g(\mathbf{x}(\tau), \mathbf{u}(\tau), \tau) d\tau \approx \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t)$ , for small  $\epsilon$

$$V^*(\mathbf{x}(t), t) = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + V^*(\mathbf{x}(t + \epsilon), t + \epsilon)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + V^*(\mathbf{x}(t + \epsilon), t + \epsilon) - V^*(\mathbf{x}(t), t)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + (V^*(\mathbf{x}(t), t) + \nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2) - V^*(\mathbf{x}(t), t)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \frac{1}{\epsilon} (\nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2))$$

$$0 = \lim_{\epsilon \rightarrow 0} \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla_{\mathbf{x}} V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) + \frac{\partial V}{\partial t} + O(\epsilon)$$

$$0 = \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)$$

# Hamilton-Jacobi-Bellman equation

Consider cost  $g_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \{g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)\} = 0$$

Terminal condition:  $V(\mathbf{x}, T) = g_N(\mathbf{x})$

Solve PDE backward in time. But generally not easy to solve ☹

# What have we just done?

- Derived the key equations describing the optimal solution to a sequential decision-making problem
  - Under some assumptions, these are necessary and sufficient conditions for optimality
- Solving exactly is hard / intractable
- Instead, perform “**approximate** dynamic programming” and **function approximators** for value function

# Can we ever hope to solve these equations exactly?!

- Yes! Under strict assumptions on the functional form of dynamics, cost, and value function.
- This is known as the linear quadratic regulator!
  - Next week!
  
- But first...HJ reachability

# Reachability analysis

- The study of the set of states a dynamical system can (and can't) reach over time given:
  - its dynamics (how it moves),
  - initial conditions,
  - possible inputs (control),
  - disturbances (uncertainty).
- Applications
  - Safety-critical control: Avoid inevitable collision states
  - Robust control: Plan a “tube path” to avoid obstacles
  - Provide guarantees: Certify safe/unsafe states before deployment

# Forward reachability (open-loop)

**Forward reachability:** The set of states a system can reach within some time using any control input

*"Starting from here, under any allowed input/disturbance, where could we end up?"*

**Open loop** because:

- Assume all possible control inputs are applied without correction along the way. I.e., no feedback.

$$\mathcal{F}(x_0, t) = \{x \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[0, t], x = \xi_{x_0, 0}^{u(\cdot)}(t)\}$$

# Backward reachability (closed-loop)

**Backward reachability:** Given a *target* set, what are all the states that could reach (or avoid) that set in the future?

*“Where must I be now to reach the target set later?”*

*“Where must I avoid now to avoid the target set later?”*

**Closed-loop** because

- At each state and time, we consider the best control to take to reach/avoid the target set

Sounds like dynamic programming...

# Backward reachable sets

## Reach case

$$\mathcal{R}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$

## Avoid case (Inevitable collision set)

$$\mathcal{A}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \forall u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$



# How to compute BRS?

- This is an optimal control problem! But we only care about whether we reach/avoid target set at the **end** of the horizon

$$\begin{aligned} & \min_{u(\cdot) \in \mathbb{U}[0, T]} \quad \int_0^T \cancel{g(x, u, t)} dt + g_T(x(T)) \\ & \text{subject to} \quad \dot{x} = f(x, u, t) \\ & \quad \quad \quad x_0 = x(0) \\ & \quad \quad \quad u \in \mathcal{U}(x), x \in \mathcal{X} \end{aligned}$$

# Define target set & value function interpretation

Consider reach case. Being inside target set is desirable

$$\mathcal{T} = \{x \in \mathcal{X} \mid g_T(x) \leq 0\}$$

Cost-to-go

$$V(x, t) = \min_{u(\cdot) \in \mathbb{U}[t, T]} g_T(x(T))$$

Do the optimal thing to get inside target set as far as possible

What about the **avoid** case?

# Going back to HJB equation (BRS)

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \{g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)\} = 0$$

Terminal condition:  $V(\mathbf{x}, T) = g_N(\mathbf{x})$

Zero running cost!

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

Solve this PDE...

# Wait a minute...what if trajectories *enter* the target set, but then escape it afterwards?

$$\mathcal{T} = \{x \in \mathcal{X} \mid g_T(x) \leq 0\} \quad V(x, t) = \min_{u(\cdot) \in \mathbb{U}[t, T]} \left( \min_{s \in [0, T]} g_T(x(s)) \right)$$

Lowest cost over horizon

# Backward reachable tube

## Reach case

$$\tilde{\mathcal{R}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \exists u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

## Avoid case (Inevitable collision set)

$$\tilde{\mathcal{A}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \forall u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

# Going back to HJB equation (BRT)

**BRS:**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

**BRT:**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min(0, \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)) = 0$$

**Idea:** Find lowest value. So we can “freeze” value ( $\frac{\partial V}{\partial t} = 0$ ) if system is moving away from target set ( $\min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \geq 0$ )

# Summary so far

- Backward reachability useful for *safety analysis*
  - Is it possible to reach/avoid a set within some time?
- We can use the HJB equation to help us solve some backward reachability problems
  - Zero running cost. Value function has a clear interpretation.
- Reach case:

**BRS:**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

**BRT:**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min(0, \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)) = 0$$

- Avoid case:  $\min_{\mathbf{u}(t) \in U(\mathbf{x}(t))}$

# What about disturbances?

- Consider disturbances in the dynamics  $\dot{x} = f(x, u, d, t)$
- We would like to be **robust** against any and all disturbances
  - Can we still reach the target set despite worst-case disturbances?
  - Can we still avoid target set despite worst-case disturbances?
- Perform reachability analysis assuming **adversarial** disturbances
  - If we want control to *reach* target, then disturbance aims to *avoid* target
  - If we want control to *avoid* target, then disturbance aims to *reach* target



# With disturbances

## Reach case

$$\mathcal{R}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \forall d(\cdot) \in \mathbb{D}[0, t], \exists u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$

$$\tilde{\mathcal{R}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \forall d(\cdot) \in \mathbb{D}[0, s], \exists u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

## Avoid case

$$\mathcal{A}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists d(\cdot) \in \mathbb{D}[0, t], \forall u(\cdot) \in \mathbb{U}[0, t] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(t) \in \mathcal{T}\}$$

$$\tilde{\mathcal{A}}(\mathcal{T}, t) = \{x_0 \in \mathcal{X} \mid \exists s \in [0, t], \exists d(\cdot) \in \mathbb{D}[0, s], \forall u(\cdot) \in \mathbb{U}[0, s] \text{ s.t. } \xi_{x_0, 0}^{u(\cdot)}(s) \in \mathcal{T}\}$$

# Ordering of control and disturbances

- Consider BRS-reach

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) = 0$$

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

or

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

- Which one should we pick?

# Max-min inequality

For any function  $f: Z \times W \rightarrow R$ ,

$$\sup_{z \in Z} \inf_{w \in W} f(z, w) \leq \inf_{w \in W} \sup_{z \in Z} f(z, w)$$

- **Interpretation**

$$\sup_{z \in Z} \inf_{w \in W} f(z, w)$$

- **Takeaway**

- Acting second has the advantage!

# HJ Reachability with disturbances

- **BRS, reach, with disturbances**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

- **BRT, reach, with disturbances**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min(0, \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \max_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t)) = 0$$

- **BRS, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \max_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \min_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t) = 0$$

- **BRT, avoid, with disturbances**

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min(0, \max_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \min_{\mathbf{d}(t) \in D(\mathbf{x}_t)} \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t)) = 0$$

# What if we have control-disturbance-affine dynamics?

- Suppose we have dynamics  $\dot{\mathbf{x}} = f_0(\mathbf{x}) + B_u(\mathbf{x})\mathbf{u} + B_d(\mathbf{x})\mathbf{d}$
- Assume control and disturbances are *bounded*
  - $\mathbf{u} \in [\mathbf{u}_{\min}, \mathbf{u}_{\max}]$ ,  $\mathbf{d} \in [\mathbf{d}_{\min}, \mathbf{d}_{\max}]$
- Consider the term  $\nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{d}(t), t)$
- Is solving for the optimal control/disturbance tractable?

# Computing HJ reachable sets

- [https://github.com/StanfordASL/hj\\_reachability](https://github.com/StanfordASL/hj_reachability)
- [https://github.com/StanfordASL/hj\\_reachability/blob/main/examples/quickstart.ipynb](https://github.com/StanfordASL/hj_reachability/blob/main/examples/quickstart.ipynb)
- 

```
dynamics = hj.systems.Air3d()

grid = hj.Grid.from_lattice_parameters_and_boundary_conditions(hj.sets.Box(np.array([-6., -10., 0.]), np.array([20., 10., 2 * np.pi])), (51, 40, 50), periodic_dims=2)

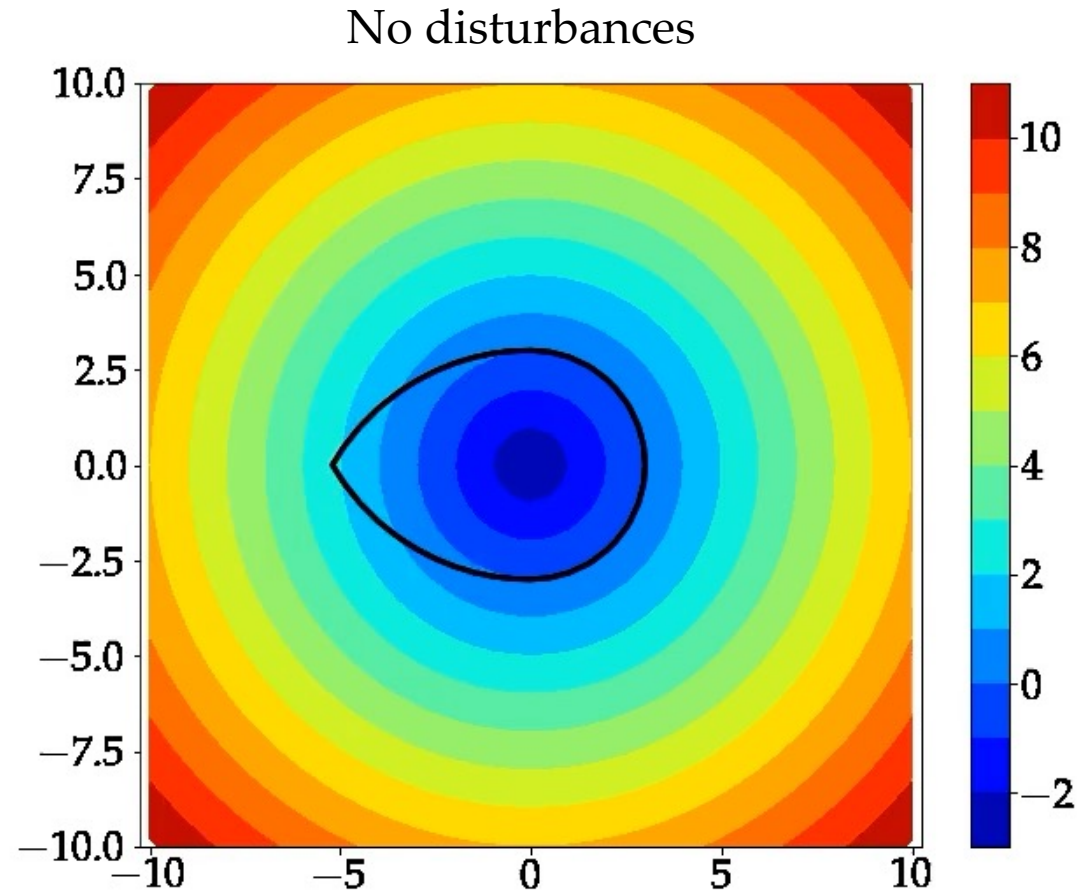
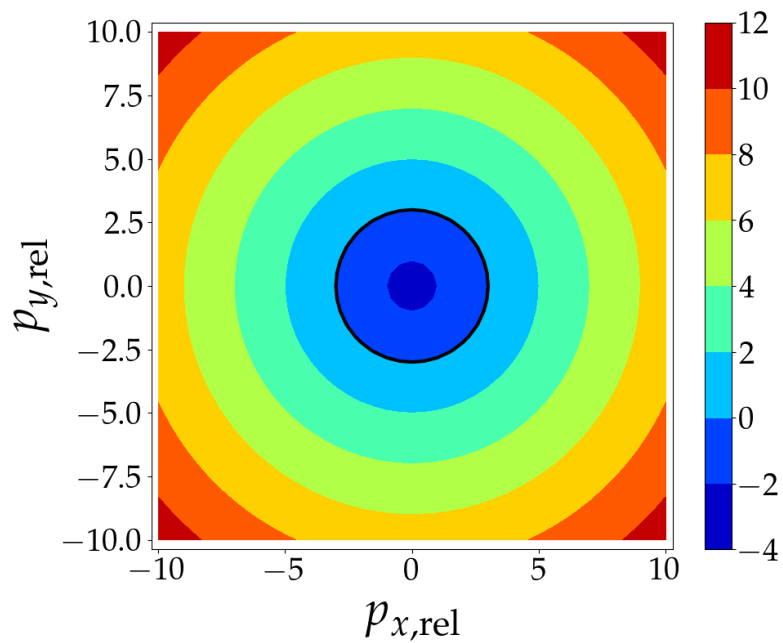
values = jnp.linalg.norm(grid.states[:, :, :2], axis=-1) - 5

solver_settings = hj.SolverSettings.with_accuracy("very_high",
hamiltonian_postprocessor=hj.solver.backwards_reachable_tube)
```

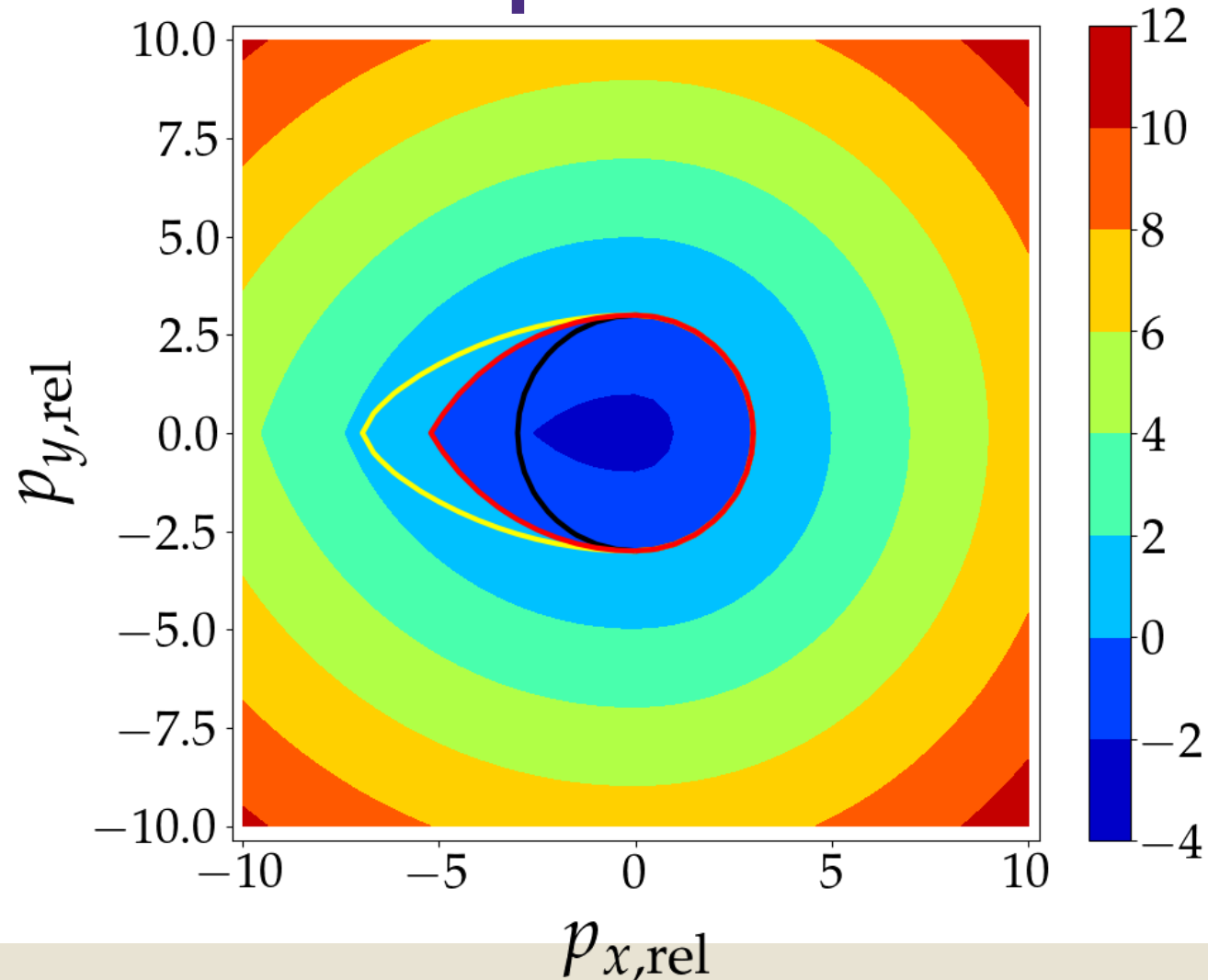
# Dubins car example (BRT, avoid)

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ u + d \end{bmatrix}$$

Constant velocity  
Can only control yaw rate



# Dubins car example with disturbances



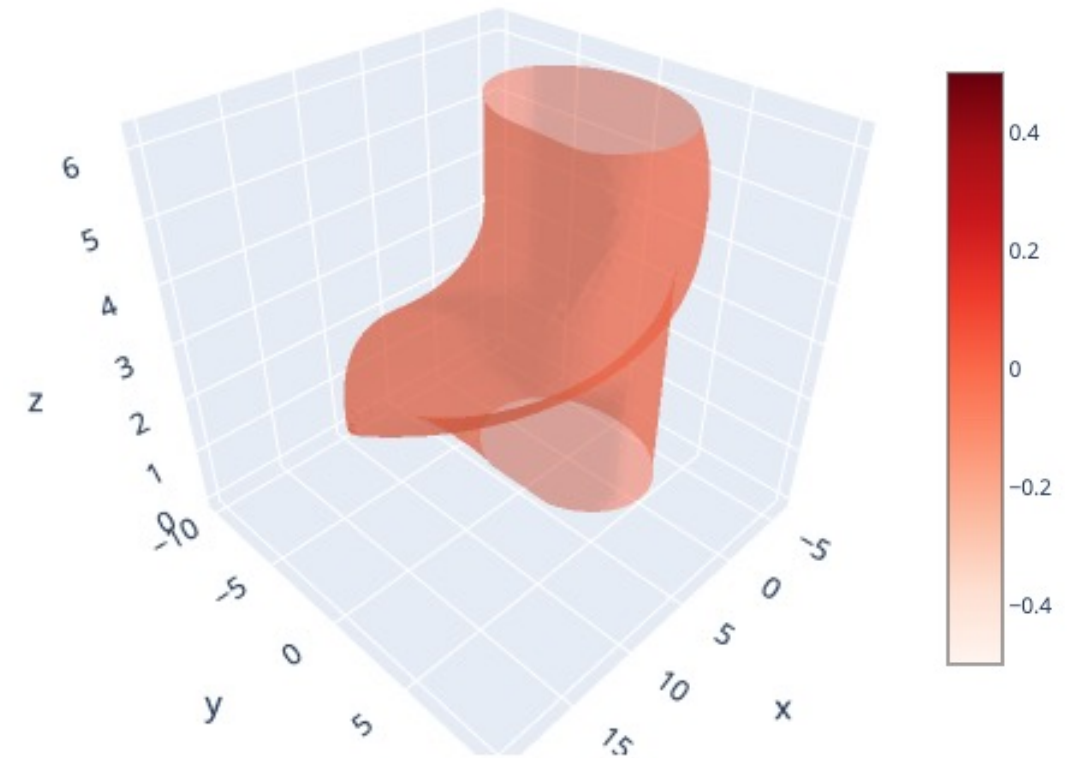
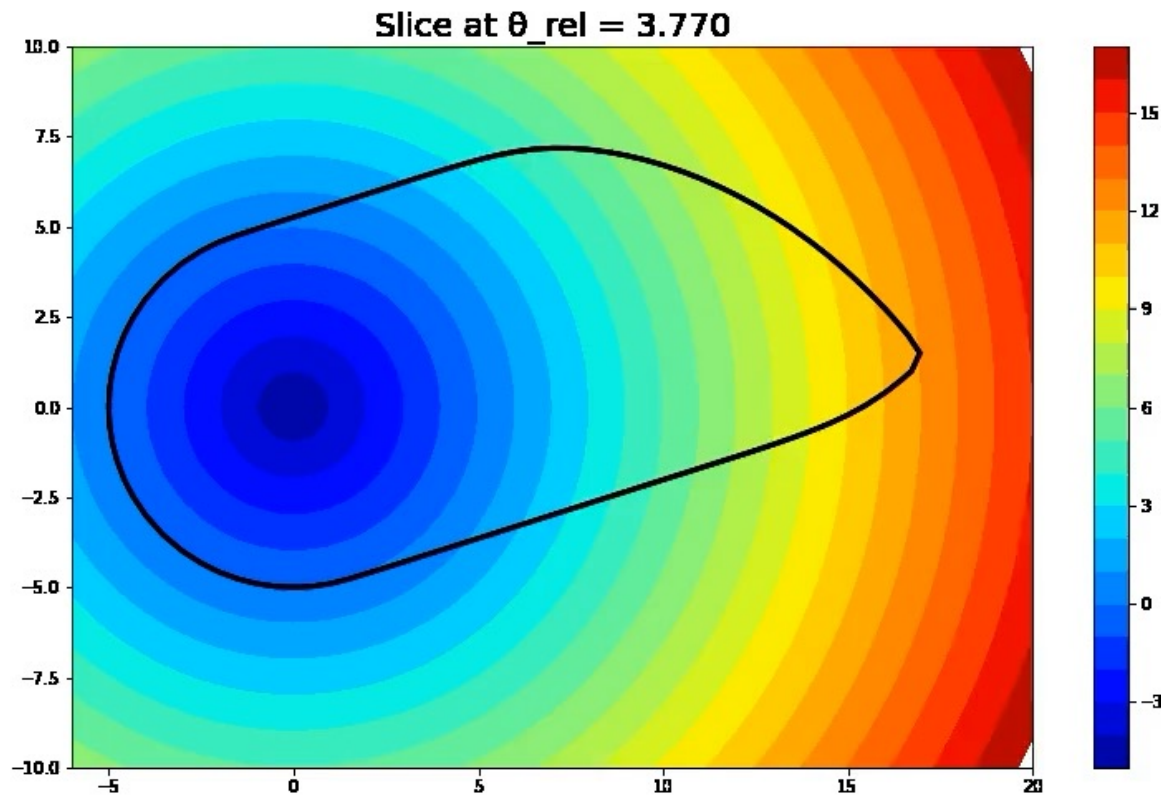


# Pursuit-evasion example

- Can consider a *relative* system between two agents & treat other agent as a disturbance
- Consider two Dubins' car agents, one is pursuing the other (i.e., cat and mouse)

$$\begin{bmatrix} \dot{x}_{\text{rel}} \\ \dot{y}_{\text{rel}} \\ \dot{\theta}_{\text{rel}} \end{bmatrix} = \begin{bmatrix} -v_a + v_b \cos \theta_{\text{rel}} + y_{\text{rel}} u_a \\ v_b \sin \theta_{\text{rel}} - x_{\text{rel}} u_a \\ u_b - u_a \end{bmatrix}$$

# Pursuit-evasion example



# What can I do with the HJ value function?

- HJ value functions are CBFs!
  - <https://arxiv.org/abs/2104.02808>
  - <https://youtu.be/9SIO-oNT7Gs>
- Least-restrictive controller
  - When  $V(x, t) \leq \epsilon$ , switch to optimal HJ policy
- Minimally-interventional safe controller
  - When  $V(x, t) \leq \epsilon$ , add HJ control constraint in tracking controller
  - <https://arxiv.org/abs/2012.03390>

# Minimally-interventional control

